COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 1 : SLO 2 — De Morgan's Theorems (Practice)
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1. Match the Expressions (Remembering)

- 1. (A + B)' → c. A'B'
- 2. (XY)' → a. X' + Y'
- 3. (A' + B')' → b. AB (since (A')'(B')' = AB)
- 4. (X'Y')' → d. X + Y
- 5. (A + B + C)' → e. A'B'C'

2. Complete the Simplification (Understanding): F = ((A + B)' . C)'

- Step 1: ((A + B)' . C)' — given expression
- Step 2: (A + B)'' + C' — De Morgan's theorem
- Step 3: (A + B) + C' — double negation
- Step 4: A + B + C' — final simplified expression

3. Simplify the Expressions (Applying)

- a. F1 = (X' + Y' + Z')' = (X')'(Y')'(Z')' = X . Y . Z
- b. F2 = (A'B)' + C = (A')' + B' + C = A + B' + C
- c. F3 = ((A + B)' + C')' = ((A + B)')' . (C')' = (A + B) . C = AC + BC

4. Find the Error (Analyzing): F = (A(B + C))'

- The error occurred in Step 2 (the student wrote A'(B + C)').
- Mistake: De Morgan's theorem states (XY)' = X' + Y'. Treating A and (B + C) as the two terms, the complement of the product must become a SUM of the complements, not a product. The student kept multiplication after negating.
- Correct simplification: F = (A(B + C))' = A' + (B + C)' = A' + B'C'

5. Create a Problem (Creating)

- a. Created expression: F = (X' + YZ)'
- b. F = (X')' . (YZ)' (De Morgan's theorem)
- F = X . (Y' + Z') (double negation + De Morgan's theorem)
- F = XY' + XZ' (distributive law)